Homework 2

Due : March 13th at 11 : 59 pm

Deliverables. Submit a single PDF of your write-up to Gradescope *HW2 Write-Up*.

Start each problem on a new page.

Honor Code

Write and sign the following statement:

> "I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

1 Paper Questions

Overview

Large Language Models (LLMs) are capable of doing amazing things! But how do we evaluate their performance? Unlike standard classification tasks, evaluation of LLM outputs involves nuance, style, and human judgment. This homework will help you explore a recent approach to evaluation: **Chatbot Arena**.

You are encouraged to skim related works, but focus on the problems, current solutions, contributions, methods, and limitations.

This assignment is not about memorizing details, but about developing the ability to *read research papers methodically.* Most research papers follow a common structure: they first motivate a **problem**, then describe **current solutions** and their limitations, followed by the **proposed solution and key insights**, the **methods** used, and finally a discussion of **limitations**. This homework is designed to help you practice and internalize this process of critical reading as you answer the questions. We have also provided pointers to relevant section that you might want to pay more attention to for each question.

- Paper: Chatbot Arena (arXiv:2403.04132)

For each question, we are looking for high-level concise answers, no need to write essays.

Chatbot Arena

a. [**Problem**] What problem is Chatbot Arena aiming to solve? Why is evaluating generative models challenging compared to classification tasks? [**Section: 1. Introduction**]

b. [**Current works**] What are current approaches to solving this problem and why are they not sufficient? [**Section: Related Works, LLM Benchmark**]

   Hint: What are the categories of LLM benchmarks outlined in the paper, and what are the limitations of each category?

c. [**Proposed solution**] What are the inputs and outputs of Chatbot Arena?

   Hint: outputs exist both per-battle and aggregated.

d. [**Key insight / contributions**] Why does the Chatbot Arena evaluation approach address the issues of prior benchmarks? What are their main contributions? (These could be methods, software artifacts, formalization of a problem, datasets, etc.)

e. [**Method details**]

   (i) What is a Bradley–Terry coefficient, and how is this used in the Chatbot Arena scoring?

   (ii) **Section 4** of the Chatbot Arena paper mentions that the model scoring system of the original Chatbot Arena interface used Elo scores instead of Bradley–Terry coefficients because they are "better for the purpose of statistical estimation." Why are Bradley–Terry coefficients more appropriate for this setting?

   Hint: Elo is an online metric, meaning that the model scores are updated after every battle, while Bradley–Terry is offline—it fits all match outcomes at once to find coefficients that best explain the entire dataset. Think about what assumptions Elo makes about the models.

Additional resource: "As a starting point, we show that the Elo score provably fails to extract the transitive component of some elementary transitive games." (Bertrand et al., 2023)

f. [**Limitations**] What are some limitations of Chatbot Arena? What are general limitations of human preference benchmarks? [**Section: 8. Discussion**]

Other Practical Resources

Machine Learning Competitions: Kaggle

Kaggle hosts many ML competitions with a tier/medal system valued by industry. Despite being competitive, there's a strong culture of code sharing that accelerates learning.

Recommended Entry-Level Competition

- Titanic: Machine Learning from Disaster

Other Relevant Competitions

- Chatbot Arena (LMSYS)
- WSDM Cup Multilingual Chatbot Arena
- LLM Detect AI-Generated Text
- LLM Classification and Fine-Tuning

2 Sum of Residuals

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

We have computed the solution w^* to the least squares regression problem:

$$w^* = \arg\min_w \frac{1}{2} \sum_{n=1}^N \left(y_n - x_n^\top w \right)^2.$$

The residual vector is then defined as:

$$r = y - Xw^*$$

(a) (*2 pts*) Use the normal equation to show that: $X^\top r = 0$.

(b) (*2 pts*) Use the presence of a bias term (encoded in the column of 1s in X) to show that the sum of the residuals is 0.

(c) (*2 pts*) Suppose we add a point (x_j, y_j) to our dataset with a very large $|y_j|$ but x_j is close to the mean of x. Predict qualitatively how w will change and explain why least squares is sensitive to such a point. How can you modify the loss to make the model more robust to outliers?

3 Weighted Linear Regression

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

Suppose we have nonnegative weights $\{\alpha_n\}_{n=1}^N \geq 0$ for each observation and our goal is to minimize the weighted least squares objective:

$$L(w) \;=\; \sum_{i=1}^{N} \alpha_i (y_i - x_i^\top w)^2.$$

In this problem, we will derive the w^* that minimizes this loss.

(a) (*2 pts*) Write the weighted least-squares loss function $L(w)$ in matrix form using the diagonal matrix:

$$A = \mathrm{diag}(\alpha_1, \ldots, \alpha_N) \in \mathbb{R}^{N \times N}.$$

Your answer should not contain any summations.

(b) (*2 pts*) Compute the gradient of the loss and express your answer in matrix form (no summations). You can do this by working with the summation form and taking partial derivatives or by using the following gradient identities:

$$\nabla_w \left(u^\top w \right) = u$$
$$\nabla_w \left(w^\top A w \right) = (A + A^\top) w$$

(c) (*3 pts*) Set the gradient equal to zero, and solve for the closed-form solution for w^* that minimizes the weighted least squares objective. in terms of X, y, and A, assuming $X^\top A X$ is invertible.

(d) (*3 pts*) Under what condition is $X^\top A X$ guaranteed to be invertible? Can we derive a simpler condition when $\alpha_i > 0$?

4 MLE vs MAP

In this problem, we explore the difference between the Maximum Likelihood Estimate (MLE) and the Bayesian Maximum A Posteriori (MAP) estimate. Suppose we have a coin with unknown bias Y. If we flip the coin we observe a Bernoulli random variable:

$$X \sim \textbf{Bernoulli}(Y) \;=\; p(X = x \mid Y) = Y^x(1-Y)^{1-x}$$

The coin will land heads ($X = 1$) with probability Y and tails ($X = 0$) with probability $1 - Y$. We flip the coin (independently) N times to collect the dataset $\mathcal{D} = \{X_n\}_{n=1}^N$. We observe the coin lands heads $N_1 = \sum_{n=1}^N X_n$ times and tails $N_0 = N - N_1$ times.

For this problem, we will also introduce a prior on the probability Y that the coin lands heads. The Beta distribution is a common prior for probabilities. The Beta distribution has the density:

$$p(y \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1}(1-y)^{\beta-1}, \quad y \in (0, 1),$$

where $\alpha > 0$ and $\beta > 0$ are parameters of the distribution. The function $B(\alpha, \beta)$ is the normalization function

$$B(\alpha, \beta) = \int_0^1 y^{\alpha-1}(1-y)^{\beta-1} dy$$

of the Beta distribution called the Beta function. As a prior we will need to pick values for α and β. These should based on our beliefs about the coin. Here we will use $\alpha = 3$ and $\beta = 3$:

(a) (*2 pts*) Write the likelihood $p(\mathcal{D}|Y)$ of the data given Y.

(b) (*2 pts*) Compute the derivative of the log-likelihood function with respect to Y and solve for the MLE estimate for Y.

(c) (*6 pts*) Write the posterior distribution $p(Y|\mathcal{D})$. Express this posterior in the form of a Beta distribution. What are the parameters of this Beta distribution in terms of α, β, N_0, and N_1?

(d) (*4 pts*) Compute the derivative of the log-posterior and solve for the MAP Estimate.

(e) (*2 pts*) In a 1 sentence, what is the difference between the MLE and MAP estimates? Note: simply putting the your solutions to the previous parts is not sufficient.

5 Estimating the Population of Grizzly Bears

An environmentalist, Amy, wants to estimate the number of grizzly bears roaming in a forest of British Columbia, Canada. She tracks $n = 20$ bears on her first visit to the forest, and marks each one with an electronic transmitter. A month later, she returns to the same forest and tracks $k = 15$ bears, and finds that only $x = 7$ of them have the transmitter on them. Assume that on each visit, she observes a uniformly random sample of bears.

Note that the numbers of bears tracked during Amy's two visits, n and k, were chosen by her. The number x of bears she found with transmitter attached is her only observation.

(a) *(2 pts)* Assuming Amy was equally likely to encounter any particular grizzly bear during her visits, what is the likelihood $\mathcal{L}(N; x)$ of the bear population N given her observation (i.e., the number of bears with transmitter observed) x?

(b) *(5 pts)* One way to estimate the bear population is to maximize the likelihood $\mathcal{L}(N; x)$. Derive the expression for the MLE estimate of the population $\hat{N}$ in terms of the number of bears tracked in both visits (parameters n, k) and the number of bears with transmitters found (observation x).

Hint: calculus may not be easy, can we use likelihood ratios instead?

(c) *(4 pts)* What is Amy's MLE estimate $\hat{N}$ of the bear population?

6 One Dimensional Mixture of Two Gaussians

Suppose we have a mixtures of two Gaussians in $\mathbb{R}$ that can be described by a pair of random variables (x, z) where x takes values in $\mathbb{R}$ and z takes value in the set $0, 1$. The joint-distribution of the pair (x, z) is given to us as follows:

$$z \sim \text{Bernoulli}(1 - \beta),$$
$$(x|z = k) \sim \mathcal{N}(\mu_k, \sigma_k^2), \quad k \in 0, 1,$$

We use θ to denote the set of all parameters $\beta, \mu_0, \sigma_0, \mu_1, \sigma_1$.

(a) *(3 pts)* Write down the expression for the joint likelihood $p(x, z = 0|\theta)$ and $p(x, z = 1|\theta)$. What is the marginal likelihood $p(x|\theta)$?

(b) *(3 pts)* What is the log-likelihood $\ell(\mathbf{x}|\theta)$? Why is this hard to optimize?

7 A Bayesian Interpretation of Lasso

Suppose you are aware that the labels $\{y_i\}_{i=1}^n$ corresponding to sample points $\{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^d$ follow the density law

$$f(y_i|\mathbf{x}_i, \mathbf{w}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2}\right)$$

where $\sigma > 0$ is a known constant and $\mathbf{w} \in \mathbb{R}^d$ is a random parameter. Suppose further that experts have told you that

- each component of $\mathbf{w}$ is independent of the others, and
- each component of $\mathbf{w}$ has the Laplace distribution with location 0 and scale being a known constant b. That is, each component $\mathbf{w}_i$ obeys the density law $f(\mathbf{w}_i) = e^{-|\mathbf{w}_i|/b}/(2b)$.

Assume the outputs $y_{i\in[n]}$ are independent from each other.

Your goal is to find the choice of parameter $\mathbf{w}$ that is *most likely* given the input-output examples $(\mathbf{x}_i, y_i)_{i\in[n]}$. This method of estimating parameters is called *maximum a posteriori* (MAP); Latin for *"maximum [odds] from what follows."*

(a) *(6 pts)* Derive the *posterior* probability density law $f(\mathbf{w}|\{(\mathbf{x}_i, y_i)\}_{i=1}^n)$ for $\mathbf{w}$ *up to a proportionality constant* by applying Bayes' Theorem and substituting for the densities $f(y_i|\mathbf{x}_i, \mathbf{w})$ and $f(\mathbf{w})$. Don't try to derive an exact expression for $f(\mathbf{w}|\{(\mathbf{x}_i, y_i)\}_{i=1}^n)$, as the denominator is very involved and irrelevant to maximum likelihood estimation.

(b) *(2 pts)* Define the log-likelihood for MAP as $\ell(\mathbf{w}) \coloneqq \ln f(\mathbf{w}|\{(\mathbf{x}_i, y_i)\}_{i=1}^n)$. Show that maximizing the MAP log-likelihood over all choices of $\mathbf{w}$ is the same as minimizing $\sum_{i=1}^n (y_i - \mathbf{w} \cdot \mathbf{x}_i)^2 + \lambda\|\mathbf{w}\|_1$ where $\|\mathbf{w}\|_1 = \sum_{j=1}^d |w_j|$ and λ is a constant. Also give a formula for λ as a function of the distribution parameters.

8 ℓ_1-regularization, ℓ_2-regularization, and Sparsity

You are given a design matrix X (whose i^{th} row is sample point $\mathbf{x}_i^\top$) and an n-vector of labels $\mathbf{y} \triangleq [y_1 \quad \dots \quad y_n]^\top$. For simplicity, assume $X^\top X = nI$. Do not add a fictitious dimension/bias term; for input $\mathbf{0}$, the output is always 0. Let $\mathbf{x}_{*i}$ denote the ith column of X.

(a) *(4 pts)* Show that the cost function for ℓ_1-regularized least squares,

$$J_1(\mathbf{w}) \coloneqq \|X\mathbf{w} - \mathbf{y}\|^2 + \lambda\|\mathbf{w}\|_1$$

(where $\lambda > 0$), can be rewritten as

$$J_1(\mathbf{w}) = \|\mathbf{y}\|^2 + \sum_{i=1}^{d} f(\mathbf{x}_{*i}, \mathbf{w}_i)$$

where $f(\cdot,\cdot)$ is a suitable function whose first argument is a vector and second argument is a scalar, and $\mathbf{w} \in \mathbb{R}^d$ is a d-dimensional weight vector.

(b) *(8 pts)* Using your solution to the previous part, derive necessary and sufficient conditions for the ith component of the optimizer $\mathbf{w}^*$ of $J_1(\cdot)$ to satisfy each of these three properties: $w_i^* > 0, w_i^* = 0$, and $w_i^* < 0$.

(c) *(3 pts)* For the optimizer $\mathbf{w}^\#$ of the ℓ_2-regularized least squares cost function $J_2(\mathbf{w}) \triangleq \|X\mathbf{w} - \mathbf{y}\|^2 + \lambda\|\mathbf{w}\|^2$ (where $\lambda > 0$), derive a necessary and sufficient condition for $\mathbf{w}_i^\# = 0$, where $\mathbf{w}_i^\#$ is the ith component of $\mathbf{w}^\#$.

(d) *(3 pts)* A vector is called *sparse* if most of its components are 0. From your solution to part (b) and (c), which of $\mathbf{w}^*$ and $\mathbf{w}^\#$ is more likely to be sparse? Why?

9 Mixed Feelings

In class, we learned about the k-means problem for clustering a distribution of data. Formally, our objective was

$$\min_{S=C_1\cup\cdots\cup C_k,\{c_1,\ldots,c_k\}} \sum_k \sum_{x\in C_k} ||x-c_k||_2^2$$

where we are simultaneously optimizing the cluster partition and our cluster centroids. We can further generalize this problem by incorporating the cluster partitions into a notion we call the "risk" R of the centroids, which measures how "bad" our centroids and assignments are. Our new objective becomes

$$\min_{c_1,\ldots,c_k} R(c_1,\ldots,c_k) = \min_{c_1,\ldots,c_k} \sum_{i=1}^{n} \min_{1\le\ell\le k} \phi(||x_i - c_\ell||_2)$$

where $\phi : \mathbb{R} \to \mathbb{R}_+$ is a continuous penalty function with minimum value $\phi(0) = 0$. This objective is a standard formulation of the generalized k-means problem (GKM). We can quickly recover the k-means objective seen in class by using $\phi(t) = t^2$.

This problem is inspired by Robust k-means: a Theoretical Revisit from NeurIPS 2016.

Breakdown of Quadratic ϕ

Assume $k = 1$ with our data $x \in \mathbb{R}$ generated from a contaminated distribution with density

$$p(x) = (1-\epsilon)\mathcal{N}(x|0,\sigma^2) + \epsilon\mathcal{C}(x|x_0,\gamma)$$

where $\epsilon \in (0,1)$ is the constant contamination fraction, $\mathcal{N}(x|0,\sigma^2)$ represents the Gaussian inliers, and $\mathcal{C}(x|x_0,\gamma)$ represents Cauchy heavy-tailed outliers (as seen in lecture). We'll investigate conditions for breakdown of our standard k-means problem using this setup.

As a reminder, here is the density for a univariate Gaussian $\mathcal{N}(x|\mu,\sigma^2)$:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

and the density for a univariate Cauchy $\mathcal{C}(x|x_0,\gamma)$:

$$p(x) = \frac{1}{\pi}\left[\frac{\gamma}{(x-x_0)^2+\gamma^2}\right]$$

(a) *(8 pts)* We define risk as the expected loss over our entire data distribution. Using this and $\phi(t) = t^2$, write the expanded integral form of the risk $R(c)$ over our distribution of x for our single centroid c. Then, prove that $R(c)$ diverges for all $c \in \mathbb{R}$ under this specific data distribution.

(b) *(4 pts)* We replace the Cauchy contamination with a finite-variance Dirac delta point mass $\delta_M(x)$ at an arbitrary distance $M > 0$. Recall that this distribution is given by

$$\delta_M(x) = \begin{cases} \infty, & x = M \\ 0, & x \neq M \end{cases}$$

Now, the density of our data is

$$p(x) = (1-\epsilon)\mathcal{N}(x|0,\sigma^2) + \epsilon\delta_M(x)$$

Find the location of the optimal centroid c^*. *Hint:* c^* minimizes risk.

Note: Since the uncontaminated Gaussian's optimal centroid is 0, your solution will demonstrate that the quadratic penalty is globally sensitive to outliers. As $M \to \infty$, the centroid shifts unboundedly away from the original centroid. As an optional exercise, you can demonstrate that in the finite data case, the contamination point is $\frac{1}{n}$, meaning a single arbitrary outlier is enough to cause the centroid estimate to diverge to infinity.

Robust k-Means

To address the sensitivity issues discovered in the quadratic penalty, we wish to introduce the robust k-means (RKM) formulation, allowing each observation x_i to take on an error term o_i, penalized by a function f_λ to encourage these errors to be zero.

We will explore how we can construct a hard-thresholding behavior where the algorithm can ignore outliers outside a fixed radius surrounding the centroid. Assume our data is generated from a continuous density $p(x) = (1-\epsilon)p_{\text{in}}(x) + \epsilon p_{\text{out}}(x)$ for $x \in \mathbb{R}^d$. We seek the optimal centroid $c^* \in \mathbb{R}^d$ and inlier weighting function $w(x) \in [0,1]$ that minimizes the expected squared error of the inliers. Additionally, this problem is constrained by the requirement that exactly $1-\alpha$ of the probability mass is assigned to the inliers.

(c) *(8 pts)* Construct a Lagrangian functional $\mathcal{L}(w, c, \lambda)$ for the weighted expected error optimization problem

$$\min_w \mathbb{E}_{x\sim p(x)}[w(x)||x-c||_2^2]$$

subject to the probability mass constraint

$$\int_{\mathbb{R}^d} w(x)p(x)\ dx = 1-\alpha$$

Then, group the terms into a single integral and optimize the Lagrangian pointwise with respect to $w(x)$ for a fixed centroid c and fixed Lagrange multiplier $\lambda > 0$. Conclude that the optimal weighting will exclude data outside some radius from the centroid.

Note: $0 \le w(x) \le 1$.

Hint: Take a look at the Wikipedia article for Lagrange multipliers if you need help getting started.

Hint: Write the constraint $g(w) = 0$ using

$$g(w) = 1-\alpha - \int_{\mathbb{R}^d} w(x)p(x)\ dx$$

Hint: Once you write the integral, can you find the optimal $w^*(x)$ for different cases of the integrand?

(d) *(5 pts)* Differentiate $\mathcal{L}$ with respect to c to find the optimal centroid c^* given the optimal weights $w^*(x)$ and write it as an expectation.

(e) *(10 pts)* Robust k-means still requires well-structured datasets where the clusters are sufficiently separated. Let's see why! Consider the one-dimensional case where $x \in \mathbb{R}$ and the data generating distribution $p(x)$ is made up of two overlapping Gaussian distributions:

$$p(x) = \frac{1}{2}\mathcal{N}(x|\mu_1, \sigma^2) + \frac{1}{2}\mathcal{N}(x|\mu_2, \sigma^2)$$

Assume $\Delta = \mu_2 - \mu_1 > 0$, $\sigma > 0$. Here, we'll attempt to find two centroids, c_1^* and c_2^*, that are optimal for our data. Write the explicit integral form for the centroid c_1^* based on your answer to the previous part, assuming $w^*(x) = \mathbb{I}_{\{|x-\mu_1|\le r\}}$ and $r > 0$. You may use $p(x)$ in your solution.

Then, prove that the bias $c_1^* - \mu_1$ is strictly positive for any finite bounding radius $r > 0$, showing that the hard boundary pulls the centroid away from the true mean.

Hint: A change of variable may help simplify your integral. Additionally, can you break up your integral into two pieces using a u-substitution? How does the Gaussian density change over squared distance?

Mixture of Gaussians

In robust k-means, we implicitly added an extra cluster to contain points past a distance threshold away from any centroid, representing a set of outliers. We also discovered a centroid bias away from the true mean. To rectify these issues, we can use a probabilistic generative model as seen in lecture: the Mixture of Gaussians (MoG).

Here, we'll modify the MoG seen in lecture to include an extra $(k+1)$-th component as a uniform distribution for the outliers. For our total data volume V, our mixture distribution with two multivariate Gaussians:

$$p(x) = \pi_1 \mathcal{N}(x|z=1, \mu_1, \Sigma) + \pi_2 \mathcal{N}(x|z=2, \mu_2, \Sigma) + \pi_{\text{out}} \mathcal{U}(x|z=\text{out})$$

where latent variable z is the cluster assignment of x, $\mathcal{U}(x) = \frac{1}{V}$ over the domain of our data, and our original constraint of $\pi_1 + \pi_2 + \pi_{\text{out}} = 1$ still applies.

(f) *(2 pts)* Let $\gamma_1(x) = p(z=1|x)$ represent the posterior probability that point x belongs to the first cluster. Write an explicit equation for $\gamma_1(x)$. For this part, you may write Gaussian densities in the form $\mathcal{N}(\ldots|\ldots)$.

(g) *(5 pts)* Prove that for an arbitrary unit vector $v \in \mathbb{R}^d$, as the point $x = \alpha v$ moves infinitely far from the centroids ($\alpha \to \infty$), the posterior probability $\gamma_1(x)$ decays to zero. Explain why the uniform background term stabilizes the denominator and find which cluster $x = \alpha v$ asymptotically belongs to.

Hint: what property of positive definite matrices can help?

10 Watch Me Flow Dat

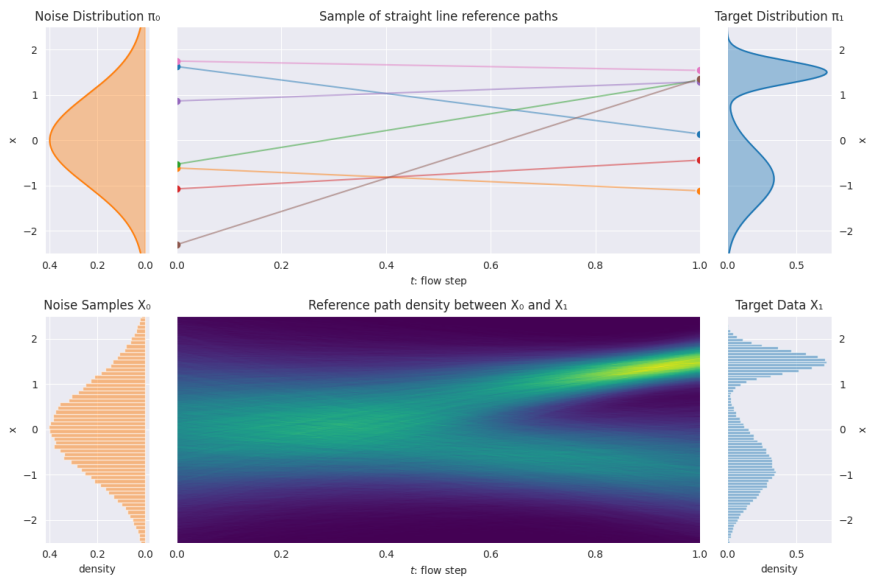

Figure 1: From Flow Matching: A visual introduction

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Another recent method of generative modeling is *flow matching* as a competitive alternative to diffusion. This problem is inspired by Flow Matching for Generative Modeling from ICLR 2023. For an optional visual introduction, we'd highly encourage A Visual Dive into Conditional Flow Matching, although all information required for this problem will be provided here.

Our goal is to map a simple prior distribution $p_0(x) = \mathcal{N}(x|0, I)$ to a complex empirical data distribution $p_1(x) \approx q(x)$. A real world example of this would be creating a mapping between pure Gaussian noise (the prior distribution) to a photographer's image dataset (the complex empirical distribution). Note the subscript here denotes time $t \in [0, 1]$ (e.g. p_1 represents the distribution at "time" $t = 1$).

We can represent the mapping between the densities with a time-dependent vector field $v_t : \mathbb{R}^d \to \mathbb{R}^d$ that assigns a velocity vector to each data point x at every $t \in [0, 1]$. The ordinary differential equation (ODE) $\frac{dx}{dt} = v_t(x)$ defines the trajectory of a point flowing between the two distributions p_0 and q. Then by integrating the ODE from $t = 0$ to $t = 1$, we can transform a point in the original distribution x_0 to a final point x_1:

$$x_1 = x_0 + \int_0^1 v_t(x(t))\ dt$$

The true marginal vector "field" $v_t(x)$ is analytically intractable, so we try to construct it using a tractable conditional probability path $p_t(x|x_1)$ targeting a specific point in the target distribution $x_1 \sim q(x_1)$. Specifically, we can create a line between any two points in the prior and target distributions $x_t = (1-t)x_0 + tx_1$. For this form, the conditional vector field guiding a point toward target x_1 is

$$u_t(x|x_1) = \frac{x_1 - x}{1 - t}$$

Take a look at the figure showing how we draw lines between points in the original and target distribution. Intuitively, you can think of $u_t(x|x_1)$ as the slope of the line (rise over run!). We treat moving along the x-axis as moving through time and moving along the y-axis as moving through space.

Given this conditional vector field, our true marginal vector field will be the expected conditional field, weighted by the posterior probability of the target data point given the current spatial state:

$$v_t(x) = \int u_t(x|x_1)p_t(x_1|x)\ dx_1$$

The Marginal Field

We first show why $v_t(x)$ has this specific integral form. For a fixed time t and fixed spatial position x, consider an arbitrary vector $v \in \mathbb{R}^d$.

(a) *(2 pt)* Write the expected squared error $J(v)$ between v and the conditional vector field $u_t(x|x_1)$ over all possible targets x_1.

Hint: what distribution should this expectation be over?

(b) *(4 pts)* Take the gradient of $J(v)$ with respect to v and prove that the minimizer v^* is the exact definition of the marginal vector field $v_t(x)$.

Equivalent Objectives

Ideally, we would fit a model $v(x, t|\theta)$ to match the marginal vector field:

$$\mathcal{L}_{\text{marginal}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,1),x\sim p_t(x)}[||v(x,t|\theta) - v_t(x)||_2^2]$$

However, this objective is intractable since we don't know the solution to $v_t(x)$. We instead optimize the Conditional Flow Matching (CFM) objective, which regresses directly against the tractable conditional fields:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,1),x_1\sim q(x_1),x\sim p_t(x|x_1)}[||v(x,t|\theta) - u_t(x|x_1)||_2^2]$$

(c) *(7 pts)* Using the definition of $v_t(x)$ from the previous parts, prove mathematically that optimizing $\mathcal{L}_{\text{CFM}}(\theta)$ is equivalent to optimizing $\mathcal{L}_{\text{marginal}}(\theta)$. Concretely, prove

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{marginal}}(\theta) + C$$

where C is a constant independent of the model parameters θ.

Hint: Bayes' theorem tells us $p_t(x|x_1)q(x_1) = p_t(x_1|x)p_t(x)$. You may find the notation $p_t(x, x_1)$ to be slightly confusing; this joint distribution should be interpreted explicitly as $p(x_t = x, x_1 = x_1)$, representing the joint probability that the process is at location x at time t and will end up at location x_1 at time 1. We use $p_t(x, x_1)$ purely for notational convenience.

MLE on Discrete Flows

We know from the previous part that optimizing $\mathcal{L}_{\text{CFM}}(\theta)$ directly matches the conditional vector fields. Now, we'll discretize the flow into a probabilistic Markov chain so we can maximize its likelihood.

Assume our time t can now be discretized into T steps of size $\Delta t = 1/T$. Let $k \in \{1, \ldots, T\}$ be the discrete step index. We observe our data as n target trajectories $\{x_0^{(i)}, \ldots, x_T^{(i)}\}_{i=1}^n$, where in each trajectory, we have $x_j^{(i)} \in \mathbb{R}^d$ generated by the true linear optimal transport conditional path $x_t = (1-t)x_0 + tx_1$.

We now model the mapping between our prior distribution and target distribution using a Markov chain, with states occuring at each discrete time-step. Using a Markov chain here is useful as we only need to model the transition probabilities efficiently and can ignore the history of a trajectory. Thus, our model will fit the transition probabilities with a multivariate Gaussian:

$$p_\theta(x_k|x_{k-1}) = \mathcal{N}\left(x_k|x_{k-1} + \Delta t \cdot v(x_{k-1}, (k-1)\Delta t|\theta), \sigma^2 I_{d\times d}\right)$$

(d) *(4 pts)* Write the log-likelihood $\ell(\theta)$ of the observed trajectories. Assume trajectories are independent, and that the process follows a first-order Markov chain assumption where the transition to state k only depends on the previous state $(k-1)$. Expand the product into a sum of multivariate Gaussian log-densities.

Note: recall the general formula for a d-dimensional multivariate Gaussian density:

$$p(x|\mu, \Sigma) = (2\pi)^{-d/2}|\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right)$$

(e) *(4 pts)* Derive the simplified MLE objective. Prove that maximizing this log-likelihood is mathematically equivalent to minimizing the squared Euclidean distance between $v(x_{k-1}^{(i)}, (k-1)\Delta t|\theta)$ and the discrete empirical velocity $\frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t}$:

$$\underset{\theta}{\operatorname{argmax}}\ \ell(\theta) = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{n}\sum_{k=1}^{T} \left\| v(x_{k-1}^{(i)}, (k-1)\Delta t|\theta) - \frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t} \right\|_2^2$$

(f) *(4 pts)* Using the linear optimal transport path definition $x_t = (1-t)x_0 + tx_1$, evaluate the discrete empirical velocity. Prove that the MLE objective derived in the previous part strictly recovers the Conditional Flow Matching objective $\mathcal{L}_{\text{CFM}}(\theta)$ when evaluated at the discrete time-steps.

Hint: Discretize the continuous optimal transport path definition using k.

Hint: Remember that the true conditional vector field is defined as

$$u_t(x_t|x_1) = \frac{x_1 - x_t}{1-t}$$